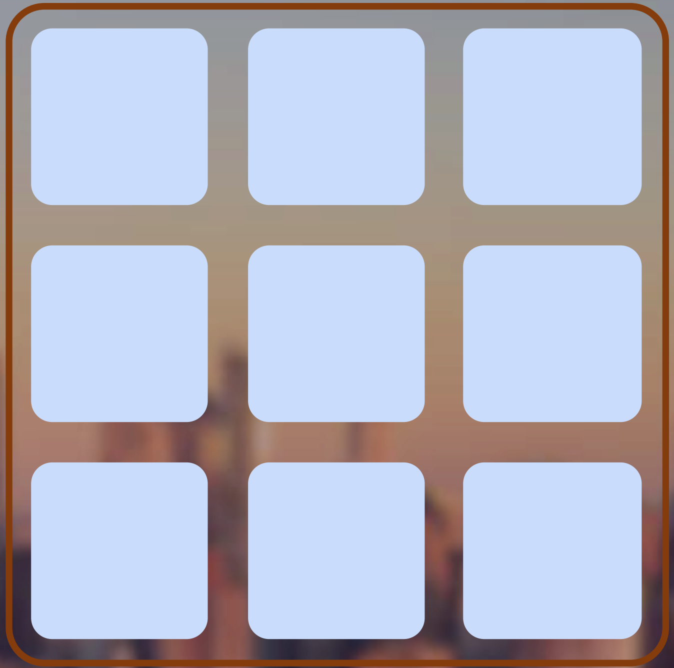
URBAN DATA ANALYSIS — FINAL PRESENTATION

Could Manhattan Have Superblocks?

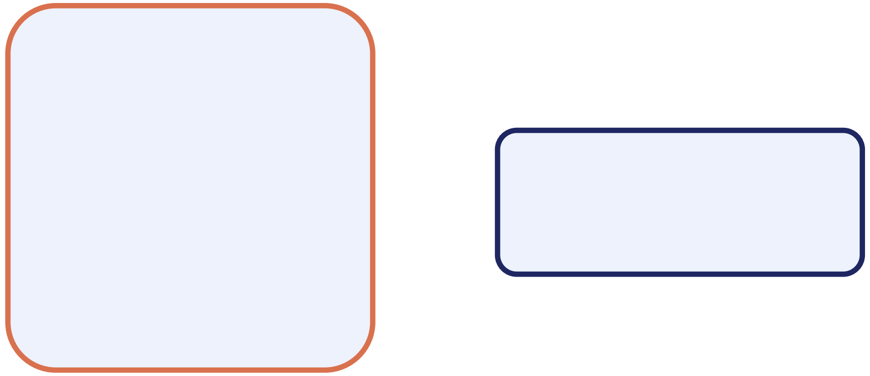
A data-driven search for Barcelona-style superblock candidate sites in the Manhattan street grid



Yerin Eun · KAIST · Fall 2026 Exchange · GitHub

A Barcelona Observation

- In the Eixample, wide chamfered intersections and traffic-calmed interior streets made the streetscape work for pedestrians and drivers alike — daily life fit within a short walk.
- The Manhattan grid looks similar on a map, but feels very different on the ground. Is that gap locked into the grid's form — or could Manhattan's streets be reallocated the way Barcelona's have been?
- Barcelona's superilles (superblocks) group 3×3 blocks and remove through-traffic from interior streets — because streets are the largest publicly-owned land reserve in a built-out city.
- Manhattan faces the same condition: buildings are fixed, but roughly a quarter of its land area is street — and it is publicly owned.



BARCELONA BLOCK

113 m × 113 m

MANHATTAN BLOCK

80 m × up to 280 m

Similar footprint, very different geometry — and, as this project asks, very different street behavior.

RESEARCH QUESTION

Where in the Manhattan street grid could Barcelona-style superblocks plausibly be implemented?

Which areas satisfy the network, traffic, and transit conditions that made superblocks feasible in Barcelona — and how much of Manhattan is structurally excluded?

WHY THIS MATTERS

Three Reasons This Question Is Worth Asking Now

1

Policy relevance and timing

NYC is already reallocating street space — Open Streets, pedestrian plazas, congestion pricing since 2025. A feasibility map speaks directly into that conversation.

2

A genuine open question

Manhattan's grid only looks like Barcelona's on the surface: elongated blocks, heavy avenue traffic and bus routes, extreme density. The answer isn't known in advance.

3

Builds on published research

Eggimann (2022, Nature Sustainability) screened 18 world cities using OpenStreetMap alone. This project localizes that method for Manhattan with real city-specific data.

DATASET

Six Real Datasets, One Screening Logic

All pulled live from public APIs — no synthetic data in the final results.

Dataset	Source	What we got
Street network	OSM / osmnx	4,634 nodes, 9,916 edges — base graph
NYC Street Centerline	NYC DCP (inkn-q76z)	30,341 segments; official street data
Traffic volume counts	NYC DOT (7ym2-wayt)	1,204 stations → 142 edges matched
Bus routes (GTFS)	MTA static feed	151 Manhattan bus shapes
Truck route network	NYC DOT (jjja-shxy)	32,939 segments
Population by block group	US Census ACS	1,292 block groups, 1.45M people

METHODOLOGY

A Four-Stage Screening Pipeline

1

Build Data Layers

Attribute the street network with hierarchy, traffic, transit, and freight data.

→

2

Generate Candidates

Enumerate superblock-sized units across the grid; tag interior vs. perimeter streets.

→

3

Exclusion Filters

Remove candidates with interior bus/truck routes or high traffic — recording what each filter removes.

→

4

Score & Rank

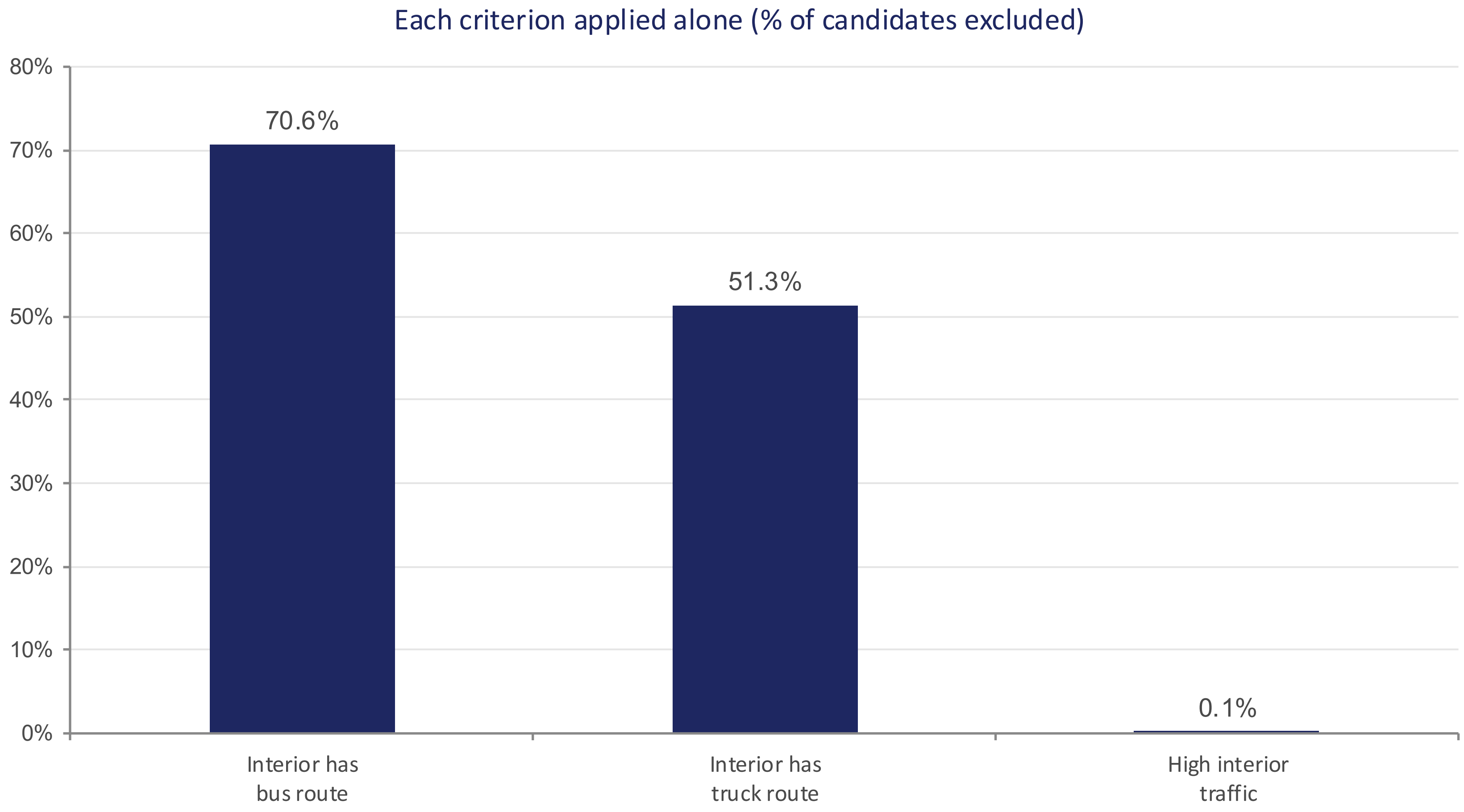
Build a normalized criteria table; rank and stress-test the surviving candidates.

Candidate unit: a sliding window bounded by an avenue pair and a run of cross streets — comparable in scale to Barcelona's 3×3 block, but elongated like Manhattan's real grid.

Bus and Truck Routes — Not Traffic — Rule Out Manhattan

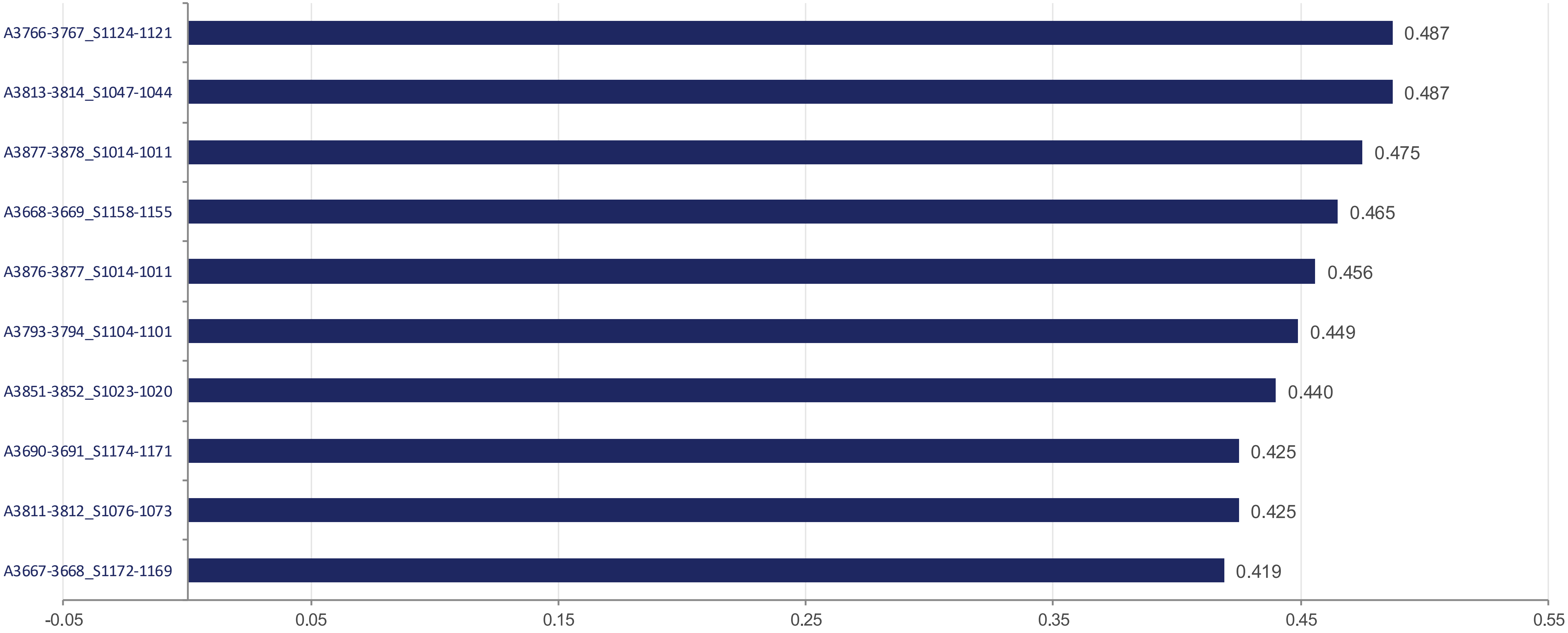
23.8%

of candidates survive
1,484 of 6,241 units



Cumulative: 6,241 → 1,836 (after bus) → 1,484 (after truck) → 1,484 (traffic adds nothing — most interior volumes were imputed near the exclusion threshold)

A Small Set of Strong Candidates — Not One Outlier



RESULTS

The Ranking Holds Up Across Different Weightings

Spearman rank correlation between suitability rankings under three weighting schemes

0.98

Baseline
vs.
Traffic-averse

0.89

Baseline
vs.
Population-first

0.87

Traffic-averse
vs.
Population-first

All three comparisons are above 0.85 — the same neighborhoods surface as top candidates regardless of whether the model leans toward penalizing traffic or rewarding population served.

RESULTS

Closing the Top Candidate: A Flag, Not a Finding

0.0%

change in average trip length
across 200 Manhattan-wide pairs

36,000 → 37,500

perimeter capacity, before →
upper-bound after closure

Why 0.0% isn't “no disruption”

- One candidate unit covers a tiny fraction of the street grid.
- 200 randomly sampled Manhattan-wide trips essentially never route through it — the test can't see a local effect.
- This is a sampling limitation, not evidence the intervention is disruption-free.
- Next step: sample trips that specifically start or end near the candidate, not uniformly across Manhattan.

DISCUSSION

Manhattan's constraint isn't traffic volume — it's the density of bus and truck routes threading through a much finer grid than Barcelona's.

23.8% of the grid structurally qualifies — and that ranking is stable across every weighting scheme we tested. Barcelona's program targets traffic volume directly; on this data, that's the criterion doing the least work in Manhattan.

DELIVERABLE

A ranked shortlist of 1,484 structurally-feasible Manhattan superblock candidates — and a quantified account of what rules the rest out.

Feasibility screening, not a design proposal. Community, political, and economic factors — including gentrification risk documented in Barcelona — sit outside the data. Traffic-volume screening is under-powered (142/8,208 edges measured) and the disruption test undercounts local effects — both discussed as limitations, not hidden.

Key reference: Eggimann, S. (2022). The potential of implementing superblocks for multifunctional street use in cities. Nature Sustainability, 5, 406–414.

Tools: osmnx, networkx, geopandas, shapely, pandas, numpy, requests, matplotlib, scipy